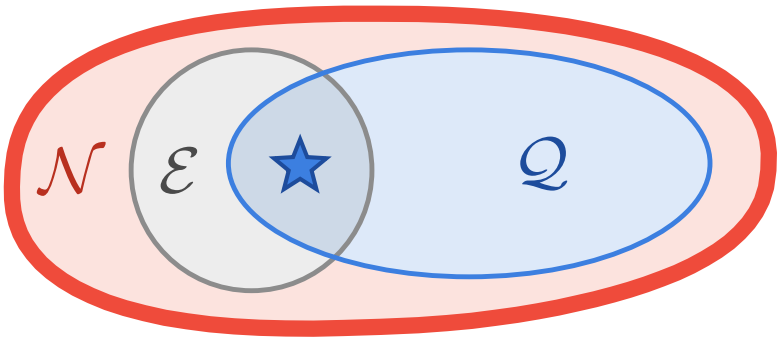
$\mathcal{N}$ quantum systems with depolarizing noise and entropy budget k

$\mathcal{Q}$ quantum reservoirs $\mathcal{E}$ measurement-free error correction

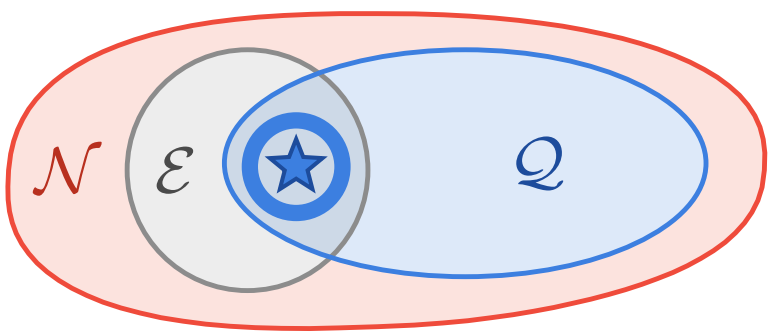
a



Upper bound, every system

$$\forall S \in \mathcal{N}: \; W(S) \leq e^{O(k)}$$

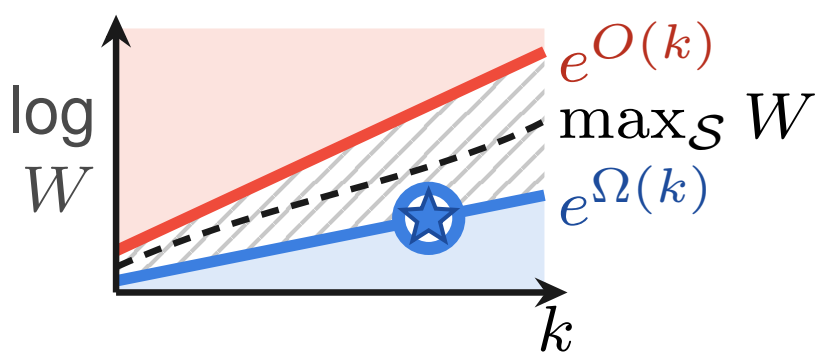
b



Lower bound, one system

$$\exists \star \in \mathcal{E} \cap \mathcal{Q}: \; W(\star) \geq e^{\Omega(k)}$$

c



Together, for $\mathcal{S} \in \{\mathcal{Q}, \mathcal{E}, \mathcal{N}\}$

$$e^{\Omega(k)} \leq \max_{S \in \mathcal{S}} W(S) \leq e^{O(k)}$$

all three maxima grow as $e^{\Theta(k)}$